EXPERIMENT 4: SECOND ORDER STEP RESPONSE OF A SERIES RLC CIRCUIT

INTRODUCTION

Consider the series RLC circuit shown in figure 1, by applying KVL around the loop for t > 0,

$$L\frac{di}{dt} + Ri + v = V_s \quad \text{.... (1)}$$

But,

$$i = C\frac{dv}{dt} \quad \text{....(2)}$$

Therefore the total solution, substitute eqn. (2) into eqn. (1)

$$\frac{d^2v}{dt^2} + \frac{R}{L}\frac{dv}{dt} + \frac{v}{LC} = \frac{V_s}{LC}$$

This solution has two components: the transient response Vt(t) and the steady-state response Vss(t); that is,

$$v(t) = v_t(t) + v_{ss}(t)$$

The transient response $V_t(t)$ for the overdamped, underdamped, and critically damped cases are:

$$v_t(t) = A_1 e^{s_1 t} + A_2 e^{s_2 t} \qquad \text{(Overdamped)}$$

$$v_t(t) = (A_1 + A_2 t)e^{-\alpha t} \qquad \text{(Critically damped)}$$

$$v_t(t) = (A_1 \cos \omega_d t + A_2 \sin \omega_d t)e^{-\alpha t} \qquad \text{(Underdamped)}$$

$$\alpha = \frac{R}{2L}, \qquad \omega_0 = \frac{1}{\sqrt{LC}}$$

$$s_1 = -\alpha + \sqrt{\alpha^2 - \omega_0^2}, \qquad s_2 = -\alpha - \sqrt{\alpha^2 - \omega_0^2}$$

$$\omega_d = \sqrt{\omega_0^2 - \alpha^2}$$

The steady-state response is the final value of V(t). In the circuit in Fig. 1, the final value of the capacitor voltage is the same as the source voltage Vs. Hence,

$$v_{ss}(t) = v(\infty) = V_s$$

Finally, the complete solutions for the overdamped, underdamped, and critically damped cases are:

$$v(t) = V_s + A_1 e^{s_1 t} + A_2 e^{s_2 t} \quad \text{(Overdamped)}$$

$$v(t) = V_s + (A_1 + A_2 t) e^{-\alpha t} \quad \text{(Critically damped)}$$

$$v(t) = V_s + (A_1 \cos \omega_d t + A_2 \sin \omega_d t) e^{-\alpha t} \quad \text{(Underdamped)}$$

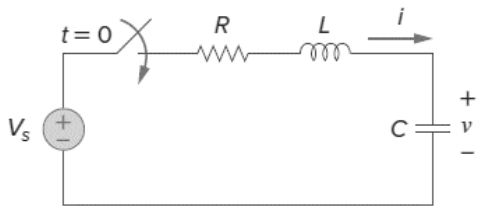


Figure 1: Step voltage applied to a series RLC circuit

OBJECTIVE

To measure the response of a series RLC circuit to a step change in driving voltage and to observe the effect that changes in one parameter, R, have upon the response.

EQUIPMENT AND COMPONENT

1. Step input voltage ,5V
2. Resistor, R (40Ω, 200Ω, 1kΩ)
3. Inductor, L (10mH)
4. Capacitor, C (1uF)

PROCEDURE

1. Wire up the circuit as shown in Figure 1. Set the value for Resistor, R = 40 Ω, inductor, L= 10mH, and capacitor, C=1uF. Set the step value to 5V for input step voltage.
2. Calculate step response parameter and fill in the data table 1.
3. Observe and record the step response curve.
4. Repeat step 1 to 3 by increasing the value of resistor to 200Ω and 1kΩ, and fill in the step response parameter in the data table 2, 3.
5. Plot all these three (3)-response curve for underdamped, overdamped and critically damped on the same graph.
6. Prove your circuit design with Multisim Simulation.

RESULT

Table 1: Underdamped response,

$\alpha < \omega_o$

Parameter	Calculated value
α	
ω_o	
ω_d	
$S_{1,2}$	
$A_1 \; and \; A_2$	
Eqn. for $V_t(t)$	
$V_t(0)$	
$V_t(\infty)$	
$V_t(1ms)$	
$V_t(2ms)$	

Table 2: Critically damped response,

$\alpha = \omega_o$

Parameter	Calculated value
α	
ω_o	
$A_1 \; and \; A_2$	
Eqn.for $V_t(t)$	
$V_t(0)$	
$V_t(\infty)$	
$V_t(0.5\,ms)$	

Table 3: Overdamped response,

$\alpha > \omega_o$

Parameter	Calculated value
α	
ω_o	
$S_{1,2}$	
$A_1 \; and \; A_2$	
Eqn.for $V_t(t)$	
$V_t(0)$	
$V_t(\infty)$	
$V_t(1ms)$	
$V_t(3ms)$	

DISCUSSION

1. Elaborate your result and findings in this experiment.
2. Compare the different between these three-step responses.
3. Which response curve are the fastest approaches the step input?
4. Show all your working calculation for Table 1, 2 and Table 3.